



UNIVERSITY OF PETROLEUM AND ENERGY
STUDIES
DEHRADUN

Modeling and Stimulation

Lab Experiment 5

MTECH-COMPUTER SCIENCE
ENGINEERING
CYBER SECURITY AND FORENSICS

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Aim

To simulate forest fire propagation using Cellular Automata and analyze the effect of environmental factors such as wind direction and intensity.

Theory :

Cellular Automata (CA) is a discrete computational model used to simulate complex systems using simple local rules. It consists of a grid of cells, where each cell changes state based on its neighbors.

In forest fire simulation:

- Each cell represents a patch of forest
- Fire spreads based on neighboring cells
- Environmental factors like wind influence spread

Cell States:

- 0 → Empty / Burnt
- 1 → Tree
- 2 → Burning

As described in the lab material , fire spreads through local interactions using a Moore neighborhood (8 neighbors).

Methodology:

2.1 Grid Initialization

- Create a 2D grid (e.g., 50×50)
- Randomly assign trees and empty spaces

2.2 Fire Initialization

- Ignite one or more cells (initial fire source)

2.3 Transition Rules

At each time step:

- Burning → Empty
- Tree → Burning (if neighbor burning + probability)
- Empty → Remains empty

2.4 Wind Influence

- Wind direction increases spread probability
- Stronger wind → faster directional spread

Algorithm:

1. Initialize grid
2. Set initial burning cell
3. For each timestep:
 - Check neighbors (Moore neighborhood)
 - Apply spread probability
 - Update grid states
4. Repeat until no fire remains
5. Visualize results

Code:

```
lab5.py > get_neighbors
1  # Author: Jigesh Sheoran
2  # Experiment 5: Cellular Automata - Forest Fire Simulation
3  # Last Modified : 10 / 03 / 2026
4
5  import numpy as np
6  import matplotlib.pyplot as plt
7  import random
8
9  # PARAMETERS
10 size = 50
11 p_tree = 0.6
12 p_fire = 0.3
13
14 wind_direction = "right"  # up, down, left, right
15 wind_strength = 0.2
16
17 steps = 30
18
19 # STATES
20 EMPTY, TREE, FIRE = 0, 1, 2
21
22 # INITIALIZE GRID
23 grid = np.random.choice([EMPTY, TREE], size*size, p=[0.4, 0.6]).reshape(size, size)
24
25 # Ignite center cell
26 grid[size//2][size//2] = FIRE
27
28 # NEIGHBOR FUNCTION
29 def get_neighbors(x, y):
30     neighbors = []
31     for dx in [-1, 0, 1]:
32         for dy in [-1, 0, 1]:
33             if dx == 0 and dy == 0:
34                 continue
35             nx, ny = x + dx, y + dy
36             if 0 <= nx < size and 0 <= ny < size:
37                 neighbors.append((nx, ny))
38     return neighbors
39
40 # UPDATE FUNCTION
41 def update(grid):
42     new_grid = grid.copy()
43
44     for i in range(size):
45         for j in range(size):
46
47             if grid[i][j] == FIRE:
48                 new_grid[i][j] = EMPTY
49
50             elif grid[i][j] == TREE:
51                 neighbors = get_neighbors(i, j)
```

```

53         for nx, ny in neighbors:
54             if grid[nx][ny] == FIRE:
55                 prob = p_fire
56
57                 # Wind effect
58                 if wind_direction == "right" and ny > j:
59                     prob += wind_strength
60                 elif wind_direction == "left" and ny < j:
61                     prob += wind_strength
62                 elif wind_direction == "up" and nx < i:
63                     prob += wind_strength
64                 elif wind_direction == "down" and nx > i:
65                     prob += wind_strength
66
67                 if random.random() < prob:
68                     new_grid[i][j] = FIRE
69                     break
70     return new_grid
71
72     # VISUALIZATION
73     burnt_counts = []
74
75     plt.figure()
76
77     for t in range(steps):
78         plt.clf()
79         plt.imshow(grid, cmap='hot')
80         plt.title(f"Forest Fire Simulation - Step {t}")
81         plt.pause(0.2)
82
83         # Count burnt cells (EMPTY)
84         burnt = np.sum(grid == EMPTY)
85         burnt_counts.append(burnt)
86
87         grid = update(grid)
88
89     plt.show()
90
91     # GRAPH: BURNT AREA VS TIME
92     plt.figure()
93     plt.plot(range(steps), burnt_counts, marker='o')
94     plt.title("Burnt Area vs Time")
95     plt.xlabel("Time Step")
96     plt.ylabel("Number of Burnt Cells")
97     plt.grid()
98
99     plt.show()

```

Output : Figure 1

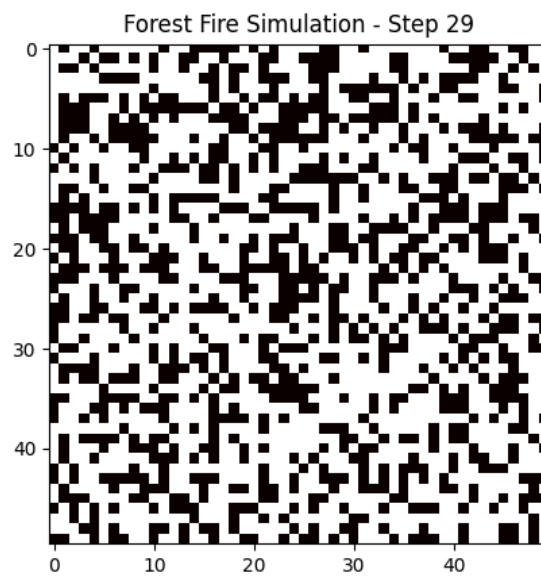
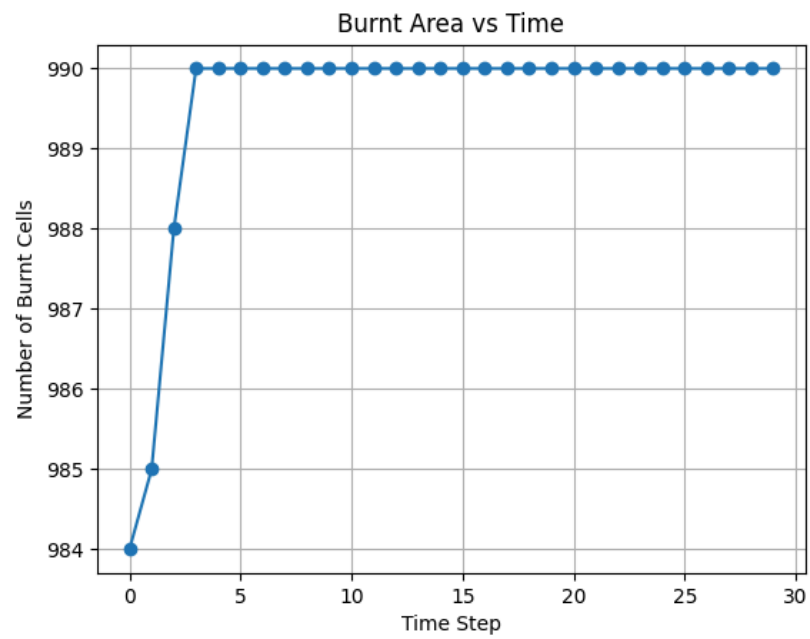


Figure 2 :



Results:

The forest fire simulation using Cellular Automata was successfully executed on a 50×50 grid. The fire propagation was visualized over multiple time steps, and the burnt area was recorded.

From the observed output:

- The fire spread rapidly across the grid within the first few time steps
- The burnt area increased sharply from approximately **984 cells to 990 cells**
- After the initial spread, the burnt area remained constant, indicating that the fire had extinguished due to lack of remaining trees

The simulation demonstrates a fast-burning scenario influenced by higher spread probability and wind effect.

Observation Table:

Time Step	Burnt Cells
0	984
1	985
2	988
3	990
4 - 29	990

Conclusion:

The experiment demonstrates that Cellular Automata can effectively model the dynamics of forest fire propagation. Based on the observed results:

- Fire spread occurred **very rapidly** in the initial time steps
- **The system quickly reached saturation, where almost all burnable cells were consumed**
- The burnt area stabilized once no further trees were available for combustion
- Environmental factors such as **wind and higher spread probability** significantly accelerated the fire propagation

Thus, the simulation highlights how local interactions and probabilistic rules can lead to rapid large-scale effects in spatial systems.

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